

Microprocessor Age **4 bit microprocessor**

4004:

- 4 bit microprocessor
- Able to address 4096 4 bit wide memory,
- Instruction set contained only 45 instructions,
- It was fabricated using a PMOS technology
- Instruction execution rate was 50 KIPs.

4040:

- Updated version of 4004.
- **Use:** microwave ovens, small control system and calculator.

8 bit microprocessor

8008: 8 bit microprocessor, 16 K bytes memory, 48 instructions.

8080: 500,000 IPS, 64 K bytes memory,

8085:

- In 1977, Intel Corporation introduced the last 8 bit microprocessor.
- Execution rate 769,230 per second.
- Main advantage was internal clock and higher clock frequency.

16 bit microprocessor

8086/8088:

- 2.5 MIPs, 1 M byte memory,
- 6 byte instruction cache or queue that prefetch a few instructions before execution

80286:

- 4 MIPs, 16 M byte memory
- Almost identical to 8086.

32 bit microprocessor

80386:

- First 32 bit microprocessor.
- 32 bit data and 32 bit memory address.
- 4 G bytes memory
- It included hardware circuitry for memory management.

80486:

- 8 K byte cache memory
- Half cycle instruction execution.

Pentium:

- 4 G byte memory, 8 K byte data cache and 8 K byte instruction cache
- Data bus 64 bit
- Multimedia execution instructions or MMX
- Dual integer processors
- The Pentium simultaneously executes two independent instructions using superscalar technology.
- Jump prediction technology of Pentium, speeds the execution of programs that include loops.
- Floating point processor processes floating point data.

Pentium Pro:

- 21 million transistors, 3 integer units, one floating point unit
- 16 K byte level 1 cache (8 K byte for data and 8 K byte for instructions) and 256 K level 2 caches
- 3 execution engines can be configured for 64 G byte memory and it is used with Windows NT operating system for server applications.

Types of Computers

Mainframe:

- The largest and most powerful computer
- They are designed to work at very high speed
- Large data words, typically 64 bits or greater
- They have massive amount of memory
- Used in military defense control, business data processing, computer graphic display.
- Example: IBM 4381

Super Computer:

- The fastest and more powerful mainframes are called Super Computer
- Example: Cray Y-MP/ 832
- Used by largest firms, government agencies and universities

Mini Computer:

- Scaled-down versions of mainframe Computer
- Runs slowly, works with smaller data word
- Does not have as much memory as mainframe
- Used in scientific research and industrial control

Micro Computer:

- Small computer
- CPU is usually a single microprocessor
- Example: Desk top, Lap top, Pam top

Microprocessor Data Type:

- Unsigned And Signed Binary Integers
- BCD (Binary Coded Decimal) Numbers
- ASCII
- Floating Point Numbers

Unsigned And Signed Binary Integers

- An unsigned binary integer has no arithmetic sign
- Example of unsigned integer is memory address
- Signed integer is represented in true form for a positive number and in two's complement form for a negative number

BCD (Binary Coded Decimal) Numbers

- Microprocessor stores BCD numbers in two forms, packed and unpacked
- The unpacked BCD number represents each BCD digit as a byte
- The packed BCD number represents two BCD digits in a byte

ASCII

- This code represents alphanumeric in a microprocessor's memory
- It also represents special symbols

Floating-Point Number

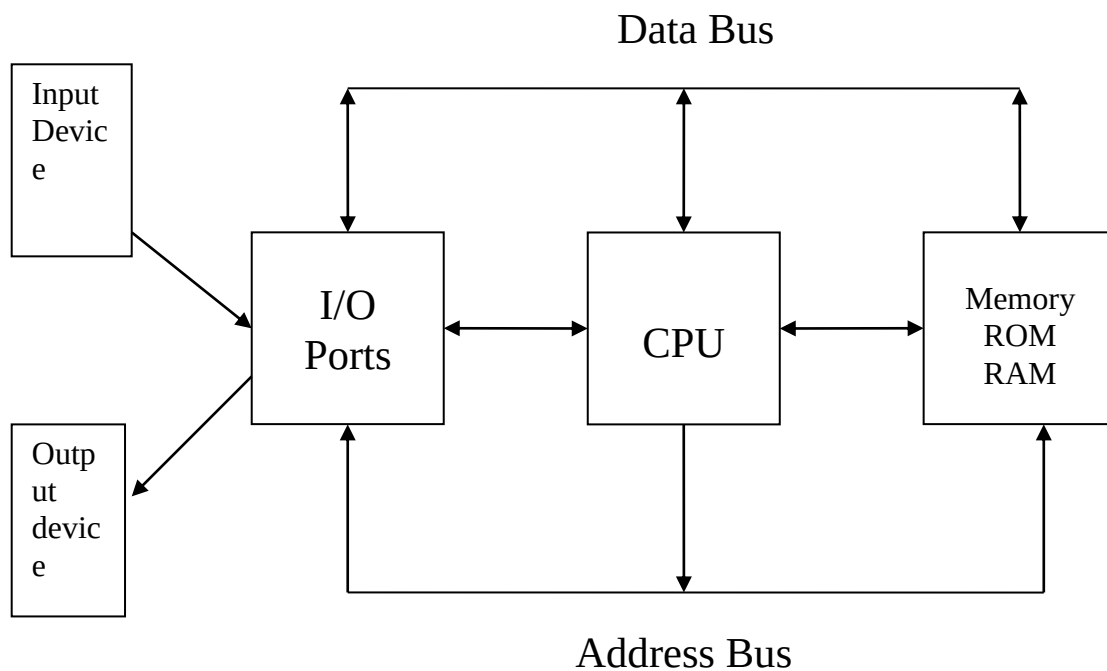
- Floating point numbers contains three components – sign, exponent and mantissa
- For the decimal value -2.5×10^{-2} , sign is negative, exponent is -2 and mantissa is 2.5
- A binary floating point number is represented as a normalized binary fraction raised to a power of 2
- Binary number is converted to a 32 bit floating point format
- Most significant bit is sign bit which is 0 for positive number and 1 for negative number
- The next 8 bits contains the bias exponent i.e. $7F_{16}$ or $+127_{10}$ is added to the exponent
- Minimum value of exponent is 0_{10} and maximum value is 255_{10}
- Remaining 23 bit represents the fractional part of the number

Example: Convert decimal number 10 to standard floating-point format.

1. $10_{10} = 1010_2 = (1.010 \times 2^3)_2$
2. Sign $S = 0$ for positive
3. Biasing exponent $= 7F_{16} + 3 = 82_{16} = 1000\ 0010_2$
4. 23 bit fraction $= 0100\ 0000\ 0000\ 0000\ 0000\ 000$
5. The floating point equivalent of 10_{10} is

S	Exponent	Fraction
0	1000 0010	0100 0000 0000 0000 0000 000

Overview of microcomputer structure and operation



Major Parts:

1. CPU
2. Memory
3. Input / Output circuitry
4. Buses:
 - a) Address bus
 - b) Data bus
 - c) Control bus

Memory:

- It stores the binary codes for the sequences of instructions
- It stores binary coded data
- Example: ROM, RAM, magnetic / optical disks

Input / Output:

- They are used to take in data from outside world or send data to the outside world
- I/O devices are connected with microprocessor through I/O ports
- Example: Keyboards, video display terminals, printers, modems

Central Processing Unit:

- It controls the operation of computer
- The CPU fetches binary-coded instructions from memory
- Decodes the instructions into a series of simple actions
- Carries out these actions in a sequence of steps
- Important components: IP, General purpose register and control bus signal generating circuits

Address Bus:

- It consists of 16, 20, 24, 32 or 36 parallel unidirectional signal lines
- On these lines the CPU sends out the address of the memory location or I/O port that is to be written to or read from
- The number of locations that the CPU can address is determined by the number of address lines

Data Bus:

- Data bus consists of 8, 16, 32 parallel bidirectional signal lines
- Many devices in the system will have their output connected to data bus, but only one device at a time will have its output enabled

Control Bus:

- The control bus consists of 4 to 10 parallel signal lines
- The CPU sends out signals on the control bus to enable the outputs of addressed memory devices or port devices
- Example of control signals: Memory read, Memory write

How to read a byte of data:

- The CPU sends the address of the desired byte on the address bus
- Then sends out a memory read signal on the control bus
- Memory read signal enables the addressed memory device to output a data word on the data bus
- The data word from memory travels along the data bus to the CPU

Memory Organization:

1. Processor memory
2. Primary or main memory
3. Secondary memory

Processor Memory:

- It refers to the microprocessor registers which are used to hold temporary results when computation is in progress
- No speed disparity between these registers and microprocessor because they are fabricated using the same technology
- Costly

Primary Memory:

- This is the storage area in which all programs are executed
- The microprocessor can directly access only those items that are stored in primary memory
- All programs and data must be within the primary memory prior to execution
- Example: ROM, RAM

Secondary Memory:

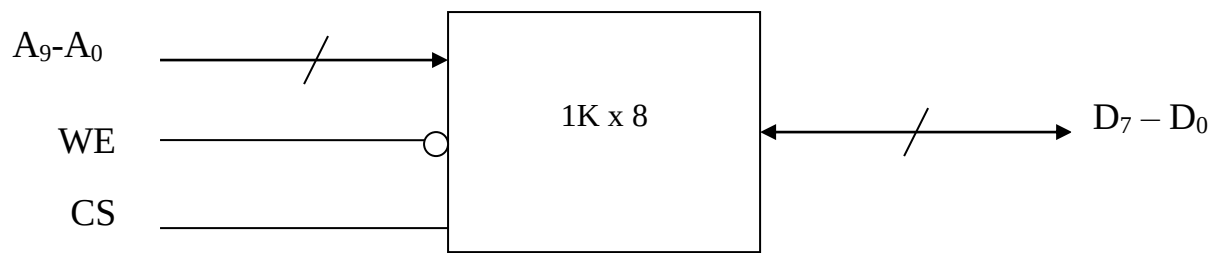
- It stores program and data in excess of main memory
- Microprocessor can not directly execute programs which are stored in secondary memory
- In order to execute these programs, the microprocessor must transfer them to its main memory by a system program called operating system
- Example: Floppy disk, Hard disk, CD etc.

Main memory array design:

Large capacity memory is realized by interconnecting several small sized memory blocks. Three techniques are frequently used.

1. Linear decoding
2. Partial decoding
3. Memory Decoding using PAL

Single RAM chip

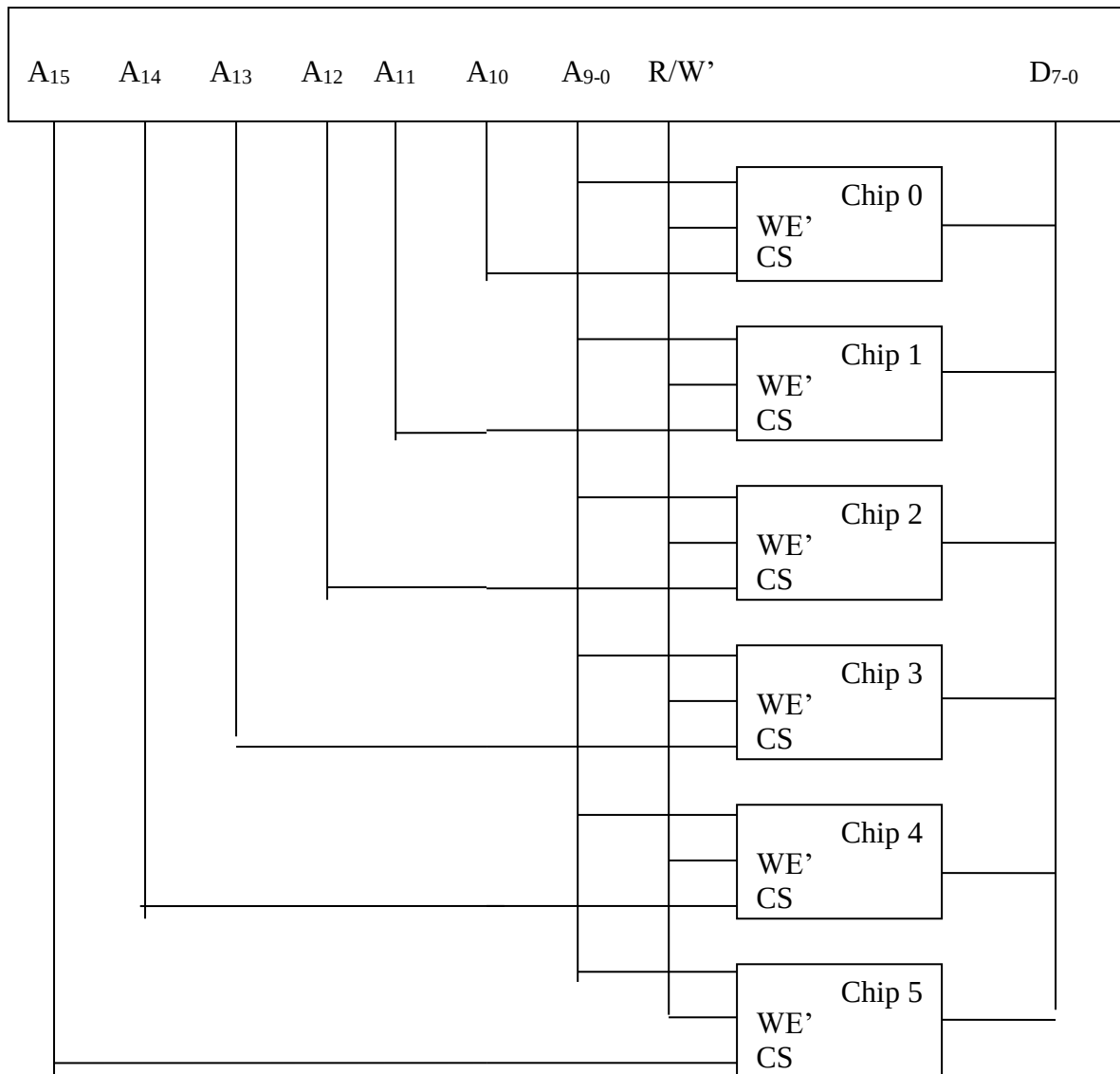


- Memory is organized as 1024 words with 8 bits / word. Each word has specific address and this is specified on 10 bit address lines $A_9 - A_0$
- The inputs and outputs are routed through 8 bit bidirectional data lines
- The operation of this chip is governed by two control inputs WE' (write enable) and CS (chip select)

CS	WE'	Mode	D ₇ – D ₀	Power
L	X	Not select	High Impedance	standby
H	L	Write	Input Bus	Active
H	H	Read	Output Bus	Active

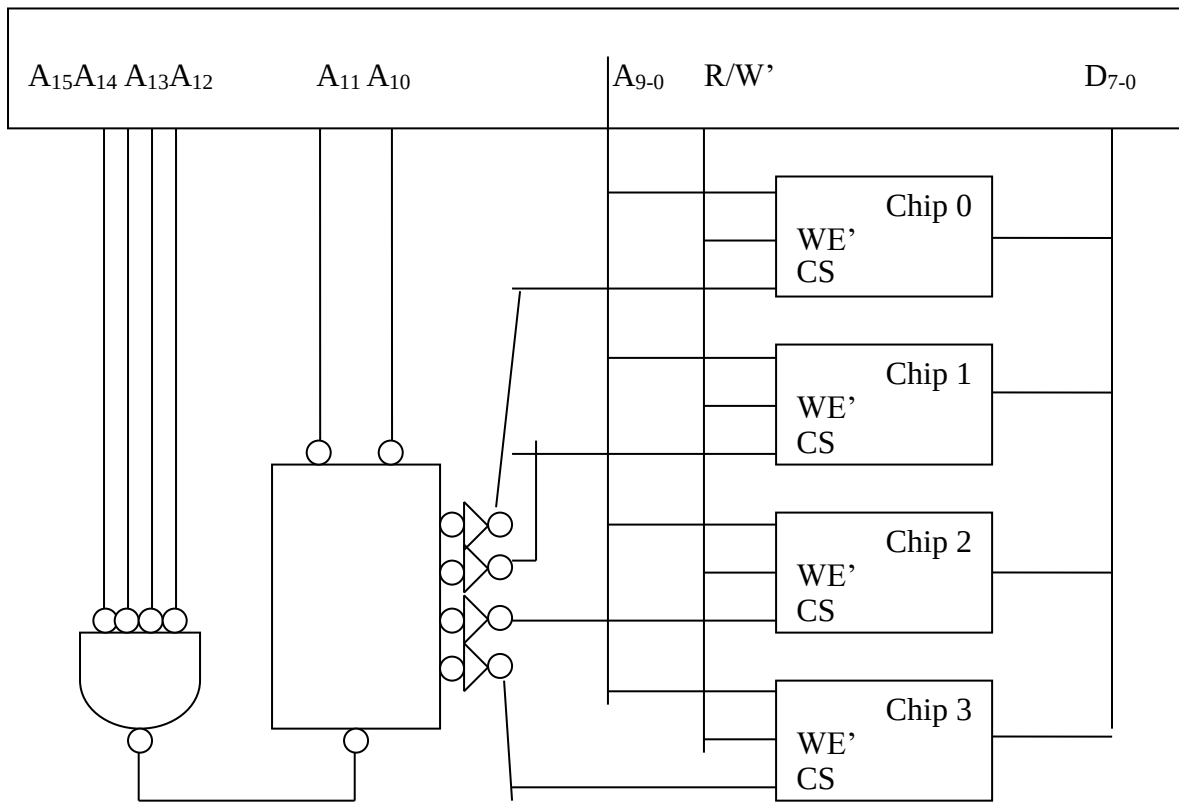
Linear decoding:

This technique uses the unused address lines of the microprocessor as chip selects for the memory chip. This method is used for small systems.



Disadvantage:

1. Wastage of address space
2. Address map is not contiguous
3. Bus conflict may occur. (If more than one chips are selected at a time)
4. Foldback (If all unused address lines are not connected as chip select)

Partial decoding:

A₁₁	A₁₀	Device
0	0	0
0	1	1
1	0	2
1	1	3

The memory system is enabled only when the lines A₁₅ through A₁₂ are zero.

Memory Decoding Using PAL